



Arnaud Gueulette

Data & AI Governance Solutions Architect

Cloud & Data Platforms | Architecture | DevSecOps & Technical Governance | AI Enablement

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PROFESSIONAL SUMMARY

Data & AI Solutions Architect combining cloud and platform engineering, data architecture, and technical governance. Hands-on with AWS, Azure, Snowflake, Collibra, Python, SQL, and Terraform, leading DEV and PROD infrastructure, CI/CD automation, platform migrations, security controls, metadata, data quality, and reference data capabilities. Experienced in translating governance and business requirements into production-grade cloud platforms and AI-ready data solutions.

PROFESSIONAL EXPERIENCE

Data & AI Governance Solutions Architect, [Datashift NV](#)

Feb 2025 – Present

- Owning the architecture and technical delivery of cloud infrastructure and platform engineering initiatives representing approximately **1.5 FTE** for [Scania CV](#)'s data governance capabilities across **DEV and PROD**, designing and operating AWS infrastructure supporting Collibra Data Quality, metadata ingestion, lineage, and technical governance.
- Designing reusable AWS infrastructure and automation for [Scania CV](#) using **EKS, Lambda, EventBridge, RDS, Terraform, Terragrunt, Docker, Kubernetes**, and **GitLab CI/CD**, standardizing deployments across environments and reducing CI/CD pipeline runtime from 90 to 30 minutes (3x faster).
- Leading infrastructure lifecycle, migration, and cost optimization activities for [Scania CV](#), including downscaling RDS in DEV and PROD to cut costs by roughly 35%, Collibra DQ metastore migration, DMS/CDC decommissioning, EventBridge integration updates, encrypted snapshot validation, and CI/CD-driven infrastructure teardown.
- Engineering cloud security and vulnerability-management controls into [Scania CV](#)'s delivery pipeline by addressing **AWS Inspector** findings, integrating automated security scanning into CI/CD, and developing policy-as-code and resource-tag enforcement patterns for IaC and CI/CD.
- Implementing and operating metadata, lineage, and data quality integrations between [Scania CV](#)'s Collibra environment and enterprise data platforms including Snowflake, Databricks, Dremio, and AWS Glue Data Catalog, covering authentication, permissions, connectivity, monitoring, and production troubleshooting.
- Providing technical leadership to a **3–4-person team** within [UCB](#)'s Data Enablement initiative while leading the design and delivery of product reference data improvements across Snowflake and external systems, resolving inconsistent identifiers and designing exact, hierarchical, and fuzzy-matching approaches to raise completeness and consistency.
- Designing and developing AI-enabled governance initiatives for [UCB](#), including Collibra MCP experimentation for natural-language metadata discovery and a policy-aware M365 Copilot agent applying **shift-left governance** by guiding compliant Collibra asset submissions at the point of creation using structured templates, JSON schemas, Excel import logic, and scenario-based testing.
- Leading the standardization of SQL-based data quality checks across multiple domains for [UCB](#), introducing structured templates, reusable query logic, and consistent validation standards.

Data & Software Engineer, [Sopra Steria Benelux](#)

Sept 2022 – Jan 2025

- **Schengen Information System, [eu-LISA](#):** enhanced and maintained border-security systems by developing, testing, and deploying improvements in PREPROD, ensuring reliability, performance, documentation quality, and alignment with EU-standard delivery requirements.
- **ISOBAR Flight Forecasting, [EUROCONTROL](#):** assessed third-party data providers through exploratory data analysis and comparative evaluation, contributing to the team's provider selection. Designed and deployed a multi-step flight-forecasting model as an online **API endpoint** to help operators optimize arrival and departure flows under varying weather conditions, reducing manual workload and supporting data-driven operational decisions.
- **Internal Projects - ESG Data Automation:** built an end-to-end pipeline to extract and process PDF-based energy-consumption invoices into the **SAS IBM Envizi ESG Suite**, improving **ESG reporting** and internal analytics.
- **Internal Projects - Analytics PoC:** delivered a **PoC** data system surfacing energy-invoice **KPIs** to support sustainability tracking, reporting, and cost-efficiency analysis.
- **Internal Projects - AI-assisted Development:** researched and documented AI-assisted development tools, including **GitHub Copilot** and prompt engineering practices, to evaluate **SDLC** productivity gains and privacy implications.
- **Internal Projects - Search:** prototyped **Elasticsearch** integration with an existing **Oracle** system to improve search performance and query expressiveness.

SELECTED ARCHITECTURE FOCUS AREAS

- **Reference Data Governance in Collibra**
Designed a practical governance model for Code Sets, Code Values, Crosswalks, lifecycle states, stewardship roles, workflows, and trusted consumption patterns, distinguishing accepted, implemented, and actively consumed reference data.
- **AI-ready Context Governance**
Developed a practical approach for using governed business, technical, operational, and policy context to support trustworthy AI agents, GenBI use cases, metadata-driven discovery, and Collibra MCP experimentation.
- **Cloud Platform & Technical Governance**
Designed and operated AWS-based DEV and PROD infrastructure, automation, security controls, and platform lifecycle processes supporting data quality, metadata, lineage, and technical governance.

EDUCATION

Master's Degree in Data Science, [Université Catholique de Louvain](#)

2020 – 2023

- Specialization in **Statistics**. Graduated **Cum Laude**.
- Thesis: Time Series Forecasting: Comparison of Box-Jenkins, Hybrid and Machine Learning Methods, comparing classical statistical methods (Box-Jenkins/ARIMA) against hybrid and machine learning approaches, including the M4-competition-winning **ES-RNN** model.

Bachelor's Degree in Business Engineering, [Université Catholique de Louvain](#)

2017 – 2020

- Specialization in **Computer Science** and **Operational Research**. Graduated **Cum Laude**.

TECHNICAL SKILLS

Data & AI Governance Platforms

Collibra Data Intelligence Platform, Collibra Data Quality, Collibra Catalog, Collibra Lineage, Collibra Workflows, BPMN, Collibra MCP

Governance Architecture & Reference Data

Stewardship Workflows, Lifecycle Management, Data Ownership, AI-ready Context Governance, Reference Data Governance (Code Sets, Code Values, Crosswalks), Metadata & Asset Models, Data Quality Frameworks & Controls, Business Glossaries, Data Lineage

Cloud & Platform Engineering

AWS S3, EC2, RDS, Glue, Glue Data Catalog, Athena, Lambda, Lake Formation, EventBridge, EKS, CloudWatch, IAM, SQS, SNS; Azure Data Lake, Data Factory, Synapse Analytics, Azure AI Services

Infrastructure as Code & DevSecOps

Terraform, Terragrunt, Docker, Kubernetes, Helm, Rancher, GitLab CI/CD, GitHub Actions, AWS Inspector, Policy-as-Code, Security Scanning

Data Engineering, APIs & AI Enablement

SQL, Python, Apache Airflow, dbt, Apache Spark, PySpark, pandas, Kafka, Debezium, FastAPI, Flask, SQLAlchemy, Pydantic, PyTest, M365 Copilot agents, MCP-based metadata discovery

Data Platforms & Analytics

Snowflake, Databricks, Dremio, PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Elasticsearch, Power BI, Apache Superset, Grafana, Kibana

CERTIFICATIONS

Collibra

- [Collibra Ranger](#)
- [Collibra Solution Architect](#)

AWS

- [AWS Certified Solutions Architect – Associate \(SAA-C03\)](#)
- [AWS Certified Data Engineer – Associate \(DEA-C01\)](#)
- [AWS Certified CloudOps Engineer – Associate \(SOA-C03\)](#)
- [AWS Certified Developer – Associate \(DVA-C02\)](#)

Soda

- [Soda Certified: Cloud Fundamentals](#)

Microsoft

- [Microsoft Certified: Azure AI Fundamentals](#)
- [Microsoft Certified: Azure Data Fundamentals](#)
- [Microsoft Certified: Azure Fundamentals](#)

SAS

- [SAS Certified Specialist: Base Programming Using SAS 9.4](#)

AWARDS

Hackathon

- [EU Defence Innovation Scheme \(EUDIS\) Hackathon 2024 - 1st Belgium, 3rd Europe](#)

LANGUAGES & PROFESSIONAL STRENGTHS

Languages: French C2/native, English C1, Dutch B1.

Strengths: stakeholder communication, problem-solving, business-driven delivery, critical thinking, continuous learning, Agile collaboration.